

Vibhas Ramani

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EDUCATION

University of Maryland

College Park, MD

Bachelor of Science in Computer Science

May 2027

Course Work: Object Oriented Programming, Discrete Structures, Intro to Computer Systems, Organization of Programming Languages, Introduction to Compilers, Introduction to Data Science, Databases, Algorithms/Data Structures, Statistics, Calculus I/II/III, Linear Algebra, Computer Architecture, Intro to Probability Theory

EXPERIENCE

NSF I-CORPS Hub Mid-Atlantic Region

June 2026 – Present

Backend Developer Intern

College Park, MD

- Engineered backend infrastructure using FastAPI, creating modular routers, RESTful APIs, and WebSocket pipelines for real-time communication
- Designed relational database systems with SQLAlchemy ORM, defining models and schemas while managing migrations through Alembic
- Built data persistence workflows using Redis and AWS S3 object storage to enable efficient retrieval of application resources
- Integrated backend endpoints with React frontend applications through JavaScript service layers and collaborated across the full-stack development pipeline

Hardware & Lumber Limited

June 2025 – August 2025

Software Engineer Intern

Kingston 11, Jamaica

- Engineered a Python + SQL data pipeline querying IBM iSeries S2K databases to compute dynamic reorder points and trigger automated stock alerts for 30+ managers nationwide
- Developed and deployed a Google Sheets extension in JavaScript that automated multi-sheet quote generation and populated Google Docs templates, streamlining customer order processing
- Saved 50+ labour hours and \$2,000 in cost by processing a backlog of over 600 quotes across retail locations.
- Built a Python tool using Pillow to batch process network-stored images and sync them to iSeries storage, reducing load times and file size by 60%
- Packaged and deployed the utility as a standalone Windows installer via Inno Setup for scalability

Children's National Hospital

February 2025 – May 2025

Full Stack Engineer Intern

College Park, MD

- Built a FastAPI backend service to integrate a brain scan ML model into the hospital platform, enabling automated medical image processing and API-based inference using Numpy and Nibabel
- Built a responsive, paginated web interface for data visualization using React, FastAPI, and Supabase
- Implemented a secure registration system with user authentication via Supabase and React Router DOM
- Collaborated in a 15-member Agile development team using Git to build and ship production React components

PROJECTS

HydroSentry AI

May 2026 — May 2026

Data Science & AI Developer

College Park, MD

- Built an ML pipeline on WPdx water infrastructure data, narrowing 73 columns to 10 key features via pandas and feature engineering
- Trained a Gradient Boosting Regressor to predict pump maintenance needs, enabling proactive repairs before failures disrupt water access
- Serialized the model and encoder, integrating inference into a Django backend for real-time predictions from user input

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, TypeScript HTML/CSS, R, Assembly, Ocaml, Rust, SQL, VBA, Postgres SQL

Frameworks & tools: Git, React, JUnit, FastAPI, MongoDB, Supabase, Pandas, NumPy, Nibabel

AWARDS

Health and Wellness Award, Gemini Hacks - November 2025

RazzleDazzle Award, Bitcamp - April 21, 2024

Nationals Second Place, World Robotics Olympiad - October 2018

OMSE Academic Excellence Award - Nominee for 2024

Creativity, Activity and Service Honor Award - Acting Director Lori Ann MacMillan-Thompson - June 2023

Citizenship Award - Guidance Counselor Hopal McClintock - June 2023